



Existence of Optimal Feedback Production Plans in Stochastic Flowshops with Limited Buffers*

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Abstract—We consider an N -machine flowshop with unreliable machines and bounds on work-in-process. Machine capacities and demand processes are finite-state Markov chains. The problem is to choose the rates of production on the machines over time to minimize the expected discounted costs of production and inventory/backlog. We show that the value function of the problem is locally Lipschitz and is a solution to a dynamic programming equation with a certain boundary condition. We provide a verification theorem, and derive the optimal feedback control policy in terms of the directional derivatives of the value function. © 1997 Elsevier Science Ltd.

1. Introduction

We consider the problem of a stochastic manufacturing system consisting of N machines in a flowshop configuration that must meet the demand for its product at a minimum cost. The stochastic nature of the system is due to the machines that are failure-prone and to uncertainty in demand. The machine capacities and demand processes are assumed to be finite-state Markov chains. The decision variables are the input rates to the machines. We take the number of parts in the buffer of the first $N - 1$ machines and the difference of the real and planned cumulative productions at the last machine, known as the surplus, as the states of the system. We consider all buffers to have storage capacities. Moreover, the number of parts in the internal buffers between any two machines should be all nonnegative, or more generally, not fall below given lower bounds. As is usual in the literature, we treat the material to be processed in the system as though it is a continuous fluid; see Chapter 2 in Sethi and Zhang (1994) and references therein for justification and details. We also assume that the processing is resumed upon a machine's repair from the point at which it was interrupted when the machine broke down. This ensures that the surplus process is continuous in time.

Our objective is to choose admissible input rates to minimize a sum of expected discounted surplus and production costs over an infinite horizon. It can be formulated as a stochastic dynamic programming problem. We show that there exists a unique optimal control under a strict convexity condition on the cost function and show that optimal control can be represented as a feedback control. Furthermore, as in Presman, Sethi and

Zhang (PSZ hereafter) (1993), we write the dynamic programming equation for the problem in terms of directional derivatives (DPEDD hereafter) for inner and boundary points and prove a verification theorem corresponding to our dynamic programming formulation.

This paper extends the work of PSZ, who only required the work-in-process to be nonnegative. Inclusion of lower and upper bound constraints on the work-in-process and upper bound on the finished good surplus represents an important feature that is usually present in real life.

There has been a substantial number of works related to the problems considered here. In view of the literature review in PSZ and Sethi and Zhang (1994), we choose not to discuss the earlier literature in this paper. Instead, we discuss the relevant research that has appeared subsequent to PSZ. Fong and Zhou (1996) have treated a two-machine flowshop with limited buffers in the context of hierarchical controls. While they are not able to show the local Lipschitz continuity of the value function as in PSZ for an n -machine flowshop with unlimited buffers or in Sethi *et al.* (1992) for a two-machine flowshop with limited internal buffer, they prove a weaker property that is sufficient for their analysis of hierarchical controls. But when it comes to optimal controls, local Lipschitz property of the value function is one of the most important things to establish. We do not know how to extend the construction procedure used in PSZ and Sethi *et al.* (1992) to allow for upper bounds on the buffer sizes. In this paper, therefore, we develop a new methodology that allows us to prove that the value function of a general N -machine flowshop is locally Lipschitz continuous. While useful in the present context, we believe that the methodology would find applications in other contexts. It therefore represents a main contribution of this paper.

The plan of the paper is as follows. In Section 2, we give a formulation of the problem and state the result on the existence and uniqueness of the optimal feedback control and establish the corresponding verification theorem. We state the main lemma required to prove that the value function is locally Lipschitz continuous. The main lemma is proved in Section 3, and the reader is referred to PSZ for proofs of the other results. Section 4 concludes the paper.

2. Problem formulation and main results

We consider a manufacturing system producing a single finished product using N machines, $M_1, \dots, M_N$, in tandem that are subject to breakdown and repair; see Fig. 1 in PSZ. We are given a stochastic process $\mathbf{k}(t) = (k_1(t), \dots, k_{N+1}(t))$ on the standard probability space $(\Omega, \mathcal{F}, P)$, where $k_n(t), n = 1, \dots, N$, is the production capacity of the n th machine M_n at time t , and $k_{N+1}(t) \equiv d(t)$ is the demand process, with constant demand representing a special case. We use $u_n(t)$ to denote the input rate to the n th machine, $n = 1, \dots, N$, and $x_n(t)$ to denote the number of parts in the buffer between the n th and $(n + 1)$ th machines, $n = 1, \dots, N - 1$. It is assumed that material is always available at M_1 . Finally, the difference between cumulative production and cumulative demand, called surplus, is denoted by $x_N(t)$. If $x_N(t) > 0$, we have finished good inventories, and if $x_N(t) < 0$, we have backlogs.

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The dynamics of the system can then be written as follows:

$$\dot{x}_n(t) = u_n(t) - u_{n+1}(t), \quad x_n(0) = x_n, \quad n = 1, \dots, N, \quad (1)$$

where $u_{N+1}(t) \equiv d(t)$. This relation can be written also in the vector form:

$$\dot{\mathbf{x}}(t) = A\mathbf{u}(t), \quad \mathbf{x}(0) = \mathbf{x}, \quad (2)$$

where $A: \mathbb{R}^{N+1} \rightarrow \mathbb{R}^N$ is the corresponding linear operator. Since the number of parts in the internal buffers cannot be negative and buffers usually have limited storage capacities, we impose the state constraints $L_n \leq x_n(t) \leq H_n, n = 1, \dots, N-1, x_N(t) \leq H_N, 0 \leq L_n < H_n \leq \infty, n = 1, \dots, N-1$, where L_n and H_n represent the lower and upper bounds on the work-in-process in the n th buffer, respectively, and H_N represents the upper bound on the finished product surplus.

To formulate the problem precisely, let $S = \prod_{n=1}^{N-1} [L_n, H_n] \times (-\infty, H_N] \subset \mathbb{R}^N$ denote the state constraint domain and let

$$U(\mathbf{k}) = \{\mathbf{u} = (u_1, \dots, u_{N+1}) : 0 \leq u_n \leq k_n, \\ n = 1, \dots, N, u_{N+1} = k_{N+1}\},$$

for $\mathbf{k} = (k_1, \dots, k_{N+1}), k_n \geq 0, n = 1, \dots, N+1$. For $\mathbf{x} \in S$, let

$$U(\mathbf{x}, \mathbf{k}) = \{\mathbf{u} : \mathbf{u} \in U(\mathbf{k}); x_n = L_n \Rightarrow u_n - u_{n+1} \geq 0, \\ n = 1, \dots, N-1, x_n = H_n \Rightarrow u_n - u_{n+1} \leq 0, \\ n = 1, \dots, N\}.$$

Let the sigma algebra $\mathcal{F}_t = \sigma\{\mathbf{k}(s) : 0 \leq s \leq t\}$. We define now the concept of admissible controls.

Definition. We say that a control $\mathbf{u}(\cdot) = (u_1(\cdot), \dots, u_{N+1}(\cdot))$ is *admissible* with respect to the initial state vector $\mathbf{x} = (x_1, \dots, x_N) \in S$ if (i) $\mathbf{u}(\cdot)$ is an $\mathcal{F}_t$ -adapted measurable process, (ii) $\mathbf{u}(t) \in U(\mathbf{k}(t))$ for all $t \geq 0$, and (iii) the corresponding state process $\mathbf{x}(t) = (x_1(t), \dots, x_N(t)) \in S$ for all $t \geq 0$.

We use $\mathcal{A}(\mathbf{x}, \mathbf{k})$ to denote the set of all admissible controls with respect to $\mathbf{x} \in S$ and $\mathbf{k}(0) = \mathbf{k}$.

Remark 1. Condition (iii) is equivalent to $\mathbf{u}(t) \in U(\mathbf{x}(t), \mathbf{k}(t)), t \geq 0$.

The problem is to find an admissible control $\mathbf{u}(\cdot)$ that minimizes the cost function:

$$J(\mathbf{x}, \mathbf{k}, \mathbf{u}(\cdot)) = E \int_0^\infty e^{-\rho t} h(\mathbf{x}(t), \mathbf{u}(t), \mathbf{k}(t)) dt, \quad (3)$$

where $h(\cdot, \cdot, \cdot)$ defines the cost of surplus and production under current value of $\mathbf{k}(t), \mathbf{k} = (k_1, \dots, k_{N+1})$ is the initial value of $\mathbf{k}(t)$, and $\rho > 0$ is the discount rate. The value function is then defined as

$$v(\mathbf{x}, \mathbf{k}) = \inf_{\mathbf{u}(\cdot) \in \mathcal{A}(\mathbf{x}, \mathbf{k})} J(\mathbf{x}, \mathbf{k}, \mathbf{u}(\cdot)). \quad (4)$$

We make the following assumptions on the random process $\mathbf{k}(t)$ and the cost function $h(\cdot, \cdot, \cdot)$:

- (A1) The capacity/demand process $\mathbf{k}(t) \in \mathcal{M}$ is a finite state Markov chain with the infinitesimal generator $Q = (q_{ij})$; see PSZ for details.
- (A2) For any given $\mathbf{k}, h(\cdot, \cdot, \mathbf{k})$ is a nonnegative jointly convex function. For all $\mathbf{x}, \mathbf{x}' \in S$ and $\mathbf{u}, \mathbf{u}' \in U(\mathbf{k}^j), j = 1, \dots, p$, there exist constants C_0 and $K_h \geq 1$ such that

$$|h(\mathbf{x}, \mathbf{u}, \mathbf{k}) - h(\mathbf{x}', \mathbf{u}', \mathbf{k})| \leq C_0(1 + |\mathbf{x}|^{K_h} + |\mathbf{x}'|^{K_h})(|\mathbf{x} - \mathbf{x}'| \\ + |\mathbf{u} - \mathbf{u}'|),$$

where $|\cdot|$ denotes an appropriate norm.

Remark 2. The restrictions on the growth rate of the cost h in (A2) is a usual assumption to ensure the existence of a solution to the dynamic programming equation to be considered later.

For the special case $L_n = 0$ and $H_n = \infty, n = 1, 2, \dots, N-1$, and $H_N = \infty$ considered in PSZ, they had assumed that the surplus cost $h(\mathbf{x}, \mathbf{u}, \mathbf{k})$ does not depend on $\mathbf{k}$, so that it can be written as $h(\mathbf{x}, \mathbf{u})$, and $h(\mathbf{x}, \mathbf{u})$ satisfies

$$|h(\mathbf{x}, \mathbf{u}) - h(\mathbf{x}', \mathbf{u}')| \leq C_0(1 + |\mathbf{x}|^{K_h} + |\mathbf{x}'|^{K_h})(|\mathbf{x} - \mathbf{x}'| \\ + C_0(|\mathbf{u} - \mathbf{u}'|), \quad (5)$$

instead of the corresponding condition in (A2). Note that all their results hold for the more general surplus cost as in (A2) above.

As in PSZ, we can formally write the DPEDD for our problem as

$$\rho v(\mathbf{x}, \mathbf{k}) = \inf_{\mathbf{u} \in U(\mathbf{x}, \mathbf{k})} \{v'_{\mathbf{u}}(\mathbf{x}, \mathbf{k}) + h(\mathbf{x}, \mathbf{u}, \mathbf{k})\} + Qv(\mathbf{x}, \mathbf{k}), \quad (6)$$

where $v'_{\mathbf{p}}(\mathbf{x}, \mathbf{k})$ is a directional derivative along the direction $\mathbf{p} \in \mathbb{R}^N$.

Theorem 1 (Existence and verification). The following results hold:

- (a) There exists an optimal control $\mathbf{u}^*(\cdot) \in \mathcal{A}(\mathbf{x}, \mathbf{k})$. Furthermore, if $h(\cdot, \cdot, \mathbf{k})$ for some $\mathbf{k}$ is strictly convex in either $\mathbf{x}$ or $\mathbf{u}$ or both, then $\mathbf{u}^*(\cdot)$ is unique, and can be represented as a feedback control, i.e., there exists a function $\mathbf{u}^*(\cdot, \cdot)$ such that for any $\mathbf{x}$, we have $\mathbf{u}^*(t) = \mathbf{u}^*(\mathbf{x}^*(t), \mathbf{k}(t)), t \geq 0$, where $\mathbf{x}^*(\cdot)$ is the optimal state process, the solution of (2) for $\mathbf{u}^*(\cdot)$ with $\mathbf{x}(0) = \mathbf{x}$.
- (b) $v(\mathbf{x}, \mathbf{k})$ satisfies equation (6) for all $\mathbf{x} \in S$.
- (c) If some continuous convex function $\tilde{v}(\mathbf{x}, \mathbf{k})$ satisfies (6) and (9) with $\mathbf{x}' = 0$, then $\tilde{v}(\mathbf{x}, \mathbf{k}) \leq v(\mathbf{x}, \mathbf{k})$. Moreover, if there exists a feedback control $\mathbf{u}(\mathbf{x}, \mathbf{k})$ providing the infimum in (6) for $\tilde{v}(\mathbf{x}, \mathbf{k})$, then $\tilde{v}(\mathbf{x}, \mathbf{k}) = v(\mathbf{x}, \mathbf{k})$ and $\mathbf{u}(\mathbf{x}, \mathbf{k})$ is an optimal feedback control.
- (d) Assume that $h(\mathbf{x}, \mathbf{u}, \mathbf{k})$ is strictly convex in $\mathbf{u}$ for each fixed $\mathbf{x}$ and $\mathbf{k}$. Let $\mathbf{u}^*(\mathbf{x}, \mathbf{k})$ denote the minimizer function of the right-hand side of (6). Then,

$$\dot{\mathbf{x}}(t) = A\mathbf{u}^*(\mathbf{x}(t), \mathbf{k}(t)), \quad \mathbf{x}(0) = \mathbf{x}$$

has a solution $\mathbf{x}^*(t)$ and $\mathbf{u}^*(t) = \mathbf{u}^*(\mathbf{x}(t), \mathbf{k}(t))$ is the optimal control.

For each $\mathbf{k} \in \mathcal{M}, \mathbf{y} \in S$, let us consider the following deterministic optimal control problem $\mathcal{P}(\mathbf{y}, \mathbf{k})$:

$$\dot{\mathbf{y}}(t) = A\mathbf{u}(t), \quad \mathbf{y}(0) = \mathbf{y}, \quad (7)$$

$$\tilde{v}(\mathbf{y}, \mathbf{k}) = \inf_{\mathbf{u}(t) \in U(\mathbf{y}(t), \mathbf{k})} \int_0^\infty e^{-\rho t} h(\mathbf{y}(t), \mathbf{u}(t), \mathbf{k}) \\ + \sum_{\mathbf{k}' \neq \mathbf{k}} q_{\mathbf{k}, \mathbf{k}'} v(\mathbf{y}(t), \mathbf{k}') dt, \quad (8)$$

where $\rho(\mathbf{k}) = \rho + \sum_{\mathbf{k}' \neq \mathbf{k}} q_{\mathbf{k}, \mathbf{k}'}$, known as the killing rate in the PDP literature (see Davis, 1993).

PSZ proved Theorem 1 in the special case when $L_n = 0, H_n = +\infty$ for $n = 1, \dots, N+1$ and $h(\mathbf{y}, \mathbf{u}, \mathbf{k})$ does not depend on $\mathbf{k}$ and satisfies (5). For this purpose, they used the reduction of initial problem to $\mathcal{P}(\mathbf{y}, \mathbf{k})$.

Remark 3. In the context of a single fluid model, Rajgopal (1994) also uses a reduction to an equivalent deterministic problem in his analysis.

The essential reduction step in PSZ was the proof of the Lipschitz property of the value function, namely,

$$|v(\mathbf{x}, \mathbf{k}) - v(\mathbf{x}', \mathbf{k})| \leq C_1(1 + |\mathbf{x}|^{K_h} + |\mathbf{x}'|^{K_h})|\mathbf{x} - \mathbf{x}'|. \quad (9)$$

To establish (9) PSZ proved an intermediate result (see Lemma 1 in PSZ). In Section 3, we provide a constructive proof of the following Main Lemma, which leads to the required Lipschitz property in this paper.

Main Lemma. Given $\mathbf{x} = (x_1, \dots, x_N), \mathbf{x}' = (x'_1, \dots, x'_N) \in S$, and $\mathbf{k} \in \mathcal{M}$, let $\mathbf{u}(\cdot) = (u_1(\cdot), \dots, u_N(\cdot), k_{N+1}(\cdot)) \in \mathcal{A}(\mathbf{x}, \mathbf{k})$ and

$\mathbf{x}(\cdot)$ be the corresponding trajectory. Then there exists $\mathbf{u}(\cdot) = (u_1(\cdot), \dots, u_N(\cdot), k_{N+1}(\cdot)) \in \mathcal{A}(\mathbf{x}, \mathbf{k})$ with the corresponding trajectory $\mathbf{x}'(\cdot)$, such that

- (a) $0 \leq u'_n(t) \leq u_n(t)$ for all $t \geq 0, n = 1, \dots, N$;
- (b) $\int_0^t |u'_n(s) - u_n(s)| ds \leq \sum_{i=1}^N |x_i - x'_i|, n = 1, \dots, N$;
- (c) $\sum_{i=1}^N |x_i(t) - x'_i(t)| \leq \sum_{i=1}^N |x_i - x'_i|$.

Armed with Main Lemma in the place of Lemma 1 of PSZ and Remark 2, the proof of Theorem 1 is absolutely similar to that in PSZ and we omit it.

3. Proof of the results

In order to prove Main Lemma, let us first consider the deterministic problem $\mathcal{P}(\mathbf{y}, \mathbf{k})$ for some initial $\mathbf{y}$ and $\mathbf{k}$ without the objective function, and deterministic functions $\mathbf{y}(t)$ and $\mathbf{u}(t), 0 \leq t < \infty$, which satisfy $\mathbf{y}(0) = \mathbf{y}_0 \in S, \mathbf{y}(t) = A\mathbf{u}(t)$, and $\mathbf{u}(t) \in U(\mathbf{y}(t), \mathbf{k})$ for the given value of $\mathbf{k}$ (it is easy to see that $\mathbf{y}(t) \in S$ for all $0 \leq t < \infty$). For any $\mathbf{y}'_0 \in S$, we will construct functions $\mathbf{y}'(t)$ and $\mathbf{u}'(t)$ which satisfy $\mathbf{y}'(0) = \mathbf{y}'_0$, $d\mathbf{y}'(t)/dt = A\mathbf{u}'(t), \mathbf{u}'(t) \in U(\mathbf{y}'(t), \mathbf{k})$ for all $0 \leq t < \infty$, and $\sum_{i=1}^N [y_i(t) - y'_i(t)]^+$ and $\sum_{i=1}^N [y_i(t) - y'_i(t)]^-$ are nonincreasing in t .

We begin our construction by defining for any $\mathbf{y}' = (y'_1, \dots, y'_N) \in S$ and $\mathbf{u} = (u_1, \dots, u_{N+1})$, the disjoint index sets

$$\begin{aligned} \alpha_+(\mathbf{y}', \mathbf{u}) &= \{n: y'_n = H_n, 1 \leq n \leq N, u_n - u_{n+1} > 0\}, \\ \alpha_-(\mathbf{y}', \mathbf{u}) &= \{n: y'_n = L_n, 1 \leq n \leq N-1, u_n - u_{n+1} \leq 0\}. \end{aligned} \quad (10)$$

Clearly, $\alpha_+(\mathbf{y}', \mathbf{u})$ is the set of indices of the components of the surplus vector $\mathbf{y}'$ that would violate the upper-bound constraint if the given control $\mathbf{u}$ were used. The set $\alpha_-(\mathbf{y}', \mathbf{u})$ is the set of indices that would either stay or violate the lower-bound constraint if the given control $\mathbf{u}$ were used.

Whenever the use of the control $\mathbf{u}$ would violate the constraint, we must modify it to avoid the violation. Two simple principles will guide our modification procedure. First, the modified control should remain nonnegative and should not exceed the given machine capacity. The way we shall ensure this is to modify the control only downward but not make it negative. Second, we shall modify the control in such a way that the new trajectory $\mathbf{y}'(\cdot)$ stays close to the given trajectory $\mathbf{y}(\cdot)$.

Let us first describe the modification when $n \in \alpha_+(\mathbf{y}', \mathbf{u})$. As is obvious from Fig. 1, reducing the upstream control u_n to equal the value of the downstream control u_{n+1} would keep the modified control between zero and the machine capacity, and ensure that the trajectory $y'_n(\cdot)$ does not violate the upper bound H_n . But, when the control u_n is reduced, it increases the rate at which the inventory in the $(n-1)$ th buffer changes. We then invoke the second principle, which suggests that we should reduce the control u_{n-1} (upstream to the $(n-1)$ th buffer) as well to preserve the difference $(u_{n-1} - u_n)$ as much as possible, i.e., without making the modified value of u_{n-1} negative.

These ideas lead us to define the modified control $\mathbf{u}^H(\mathbf{y}', \mathbf{u}) = (u_1^H(\mathbf{y}', \mathbf{u}), \dots, u_{N+1}^H(\mathbf{y}', \mathbf{u}))$ by the following backward induction:

$$\begin{aligned} u_{N+1}^H(\mathbf{y}', \mathbf{u}) &= u_{N+1}, \\ u_n^H(\mathbf{y}', \mathbf{u}) &= \begin{cases} u_{n+1}^H(\mathbf{y}', \mathbf{u}) & \text{if } n \in \alpha_+(\mathbf{y}', \mathbf{u}), \\ [u_n - u_{n+1} + u_{n+1}^H(\mathbf{y}', \mathbf{u})]^+ & \text{if } n \notin \alpha_+(\mathbf{y}', \mathbf{u}). \end{cases} \end{aligned} \quad (11)$$

Similar arguments lead us to a forward induction procedure for modifying the control u_n when $n \in \alpha_-(\mathbf{y}', \mathbf{k})$. The procedure results in $\mathbf{u}^L(\mathbf{y}', \mathbf{u}) = (u_1^L(\mathbf{y}', \mathbf{u}), \dots, u_{N+1}^L(\mathbf{y}', \mathbf{u}))$ defined as follows:

$$\begin{aligned} u_1^L(\mathbf{y}', \mathbf{u}) &= u_1, \\ u_{n+1}^L(\mathbf{y}', \mathbf{u}) &= \begin{cases} u_n^L(\mathbf{y}', \mathbf{u}) & \text{if } n \in \alpha_-(\mathbf{y}', \mathbf{u}), \\ [u_{n+1} - u_n + u_n^L(\mathbf{y}', \mathbf{u})]^+ & \text{if } n \notin \alpha_-(\mathbf{y}', \mathbf{u}), n < N, \\ u_{N+1} & \text{if } n = N. \end{cases} \end{aligned} \quad (12)$$

It should be obvious that if $\mathbf{u}$ violates no state constraints, then no modification is required. Furthermore, we should note that if using $\mathbf{u}$ would violate only the upper bound (respectively, lower

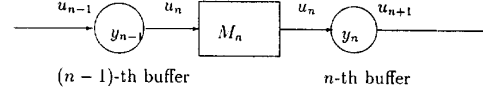


Fig. 1. Upstream and downstream buffers of machine M_n .

bound) constraints, then $\mathbf{u}^H(\mathbf{y}', \mathbf{u})$ (respectively, $\mathbf{u}^L(\mathbf{y}', \mathbf{u})$) would provide the modified control we are seeking. But, in general, upper bound constraints may be violated at some buffers and lower bound constraints at some others. Then it seems reasonable that we first modify the control $\mathbf{u}$ according to the backward procedure (11) designed to avoid violations of the upper bound constraints, and then further modify the resulting control $\mathbf{u}^H$ in turn according to the forward procedure (12) designed to avoid violations of the lower bound constraints. Thus, we consider the following control

$$\mathbf{u}'(\mathbf{y}', \mathbf{u}) = \mathbf{u}^L(\mathbf{y}', \mathbf{u}^H(\mathbf{y}', \mathbf{u})), \quad (13)$$

and the resulting trajectory obtained from solving the differential equation

$$\frac{d}{dt} \mathbf{y}'(t) = A\mathbf{u}'(\mathbf{y}'(t), \mathbf{u}(t)), \quad \mathbf{y}'(0) = \mathbf{y}'_0. \quad (14)$$

Remark 4. In order to prove their Lemma 1 (corresponding to Main Lemma of this paper), PSZ constructed the stochastic control process $\mathbf{u}'(\cdot) \in \mathcal{A}(\mathbf{x}', \mathbf{k})$, for the new initial condition $\mathbf{x}'$, given $\mathbf{x}, \mathbf{k}$ and $\mathbf{u}(\cdot) \in \mathcal{A}(\mathbf{x}, \mathbf{k})$. Here we construct the transformation (13), which is local in nature and independent of the machine state $\mathbf{k}$, and which depends only on the new current value (or value at time t) of $\mathbf{y}'$ and the time- t value of the given control $\mathbf{u}$. While this transformation resembles a projection of $\mathbf{u}$ onto the set of admissible directions, it is not the same. Moreover, we are unable to prove Lemma 2 below by the projection method.

We can state and prove the following results concerning the deterministic control defined by (13) and the deterministic trajectory defined by (14). The first lemma says that the modified control $\mathbf{u}'$ is nonnegative and is no larger than the given control $\mathbf{u}$. The second says that the corresponding trajectory $\mathbf{y}'(t)$ stays within the required upper and lower bounds for $t \in [0, \infty)$. Intuitively, it is not difficult to see why these results should follow from our construction. The second lemma also says that the deterministic trajectory $\mathbf{y}'(\cdot)$, emanating from the given initial condition $\mathbf{y}$ and the control $\mathbf{u}(\cdot)$, and the new trajectory $\mathbf{y}'(\cdot)$ do not stray too far away from one another from where they begin at time zero.

Lemma 1. If $\mathbf{u} \in U(\mathbf{y}, \mathbf{k})$, then

- (i) $0 \leq u'_n(\mathbf{y}', \mathbf{u}) \leq u_n, 1 \leq n \leq N$;
- (ii) $\mathbf{u}'(\mathbf{y}', \mathbf{u}) \in U(\mathbf{y}', \mathbf{k})$.

Proof. In order to prove (i), it suffices in view of $u_{N+1}^H(\mathbf{y}', \mathbf{u}) = u_{N+1}$ to show the following:

- (H₁) $0 \leq u_n^H(\mathbf{y}', \mathbf{u}) \leq u_n, 1 \leq n \leq N+1$, and
- (L₁) $0 \leq u_n^L(\mathbf{y}', \mathbf{u}) \leq u_n, 1 \leq n \leq N+1$.

To prove (H₁), we use the induction argument with the knowledge that $0 \leq (a+b)^+ \leq a$ for $a = u_n \geq 0$ and $b = u_{n+1}^H(\mathbf{y}', \mathbf{u}) - u_{n+1} \leq 0$. We can prove (L₁) analogously.

With (i) in hand, the proof of (ii) will be complete once we show that

- (H₂) $u'_n(\mathbf{y}', \mathbf{u}) - u'_{n+1}(\mathbf{y}', \mathbf{u}) \leq 0$ for $n \in \alpha_+(\mathbf{y}', \mathbf{u})$, and
- (L₂) $u'_n(\mathbf{y}', \mathbf{u}) - u'_{n+1}(\mathbf{y}', \mathbf{u}) \geq 0$ for $n \in \alpha_-(\mathbf{y}', \mathbf{u})$.

In order to prove (H₂), note that if $n \in \alpha_+(\mathbf{y}', \mathbf{u})$, then $y'_n = H_n$ and, therefore, $n \notin \alpha_-(\mathbf{y}', \mathbf{u}^H(\mathbf{y}', \mathbf{u}))$. According to (13),

$$\begin{aligned} u'_{n+1}(\mathbf{y}', \mathbf{u}) &= u_{n+1}^L(\mathbf{y}', \mathbf{u}^H(\mathbf{y}', \mathbf{u})) \\ &= [u_{n+1}^H(\mathbf{y}', \mathbf{u}) - u_n^H(\mathbf{y}', \mathbf{u}) + u_n^L(\mathbf{y}', \mathbf{u}^H(\mathbf{y}', \mathbf{u}))]^+ \\ &= [u_{n+1}^H(\mathbf{y}', \mathbf{u}) - u_n^H(\mathbf{y}', \mathbf{u}) + u'_n(\mathbf{y}', \mathbf{u})]^+. \end{aligned}$$

But $n \in \alpha_-(y', u)$ and (11) imply that $u_{n+1}^H(y', u) = u_n^H(y', u)$. Thus,

$$u'_{n+1}(y', u) - u'_n(y', u) = 0.$$

This completes the proof of (H_2) . The proof of (L_2) is similar. $\square$

Lemma 2. Equation (14) has a solution $y'(\cdot)$ such that $y'(t) \in S$, $0 \leq t < \infty$. Moreover, $y'(\cdot)$ is such that $\sum_{i=1}^N [y_i(t) - y'_i(t)]^+$ and $\sum_{i=1}^N [y_i(t) - y'_i(t)]^-$ are nonincreasing in t .

Proof. Let us first consider the case when $u(\cdot)$ is a piecewise-constant, right-continuous deterministic function. Then from the construction of the feedback control $u'(y', u)$ and Lemma 1(ii), it is easy to see that a solution $y(\cdot)$ to (14) exists with $y(t) \in S$ for $0 \leq t < \infty$.

To prove the second part, we use (14) to write

$$\frac{d}{dt}(y(t) - y'(t)) = A(u(t) - u'(y'(t), u(t))).$$

For $y, y' \in S$, let us define the following partition of $\{1, \dots, N\}$:

$$\begin{aligned} \alpha_+(y, y', u) &= \{n: y_n - y'_n < 0 \text{ or } [y_n - y'_n \\ &= 0, (A(u - u'(y', u)))_n < 0]\}, \\ \alpha_-(y, y', u) &= \{n: y_n - y'_n > 0 \text{ or } [y_n - y'_n \\ &= 0, (A(u - u'(y', u)))_n \geq 0]\}, \end{aligned} \quad (15)$$

where we note that

$$(A(u - u'(y', u)))_n = u_n - u_{n+1} - (u'_n(y', u) - u'_{n+1}(y', u)). \quad (16)$$

As we shall see next, definition (15) of the new index sets allows us to consider the negative and positive summands of $\sum_{i=1}^N (y_i(t) - y'_i(t))$ separately.

We now consider the sum of the negative differences. It is easy to see that for the piecewise-constant, right-continuous control $u(\cdot)$, a fixed t , and a sufficiently small value of s , we have

$$\begin{aligned} &\sum_{n=1}^N [y_n(t+s) - y'_n(t+s)]^- \\ &= \sum_{n \in \alpha_-(y(t), y'(t), u(t))} [y'_n(t+s) - y_n(t+s)] \\ &= \sum_{n=1}^N [y_n(t) - y'_n(t)]^- \\ &\quad - s \sum_{n \in \alpha_-(y(t), y'(t), u(t))} (A(u(t) - u'(y'(t), u(t))))_n. \end{aligned}$$

It is clear that for $\sum_{n=1}^N [y_n(t+s) - y'_n(t+s)]^-$ to be nonincreasing, it is enough that for any y, y' , and u , we have

$$\sum_{n \in \alpha_-(y, y', u)} (A(u - u'(y', u)))_n \geq 0. \quad (17)$$

To show (17), we need to explore the relationship between the index sets defined in (10) and (15). Since $u_n - u_{n+1} > 0$ and $y'_n = H_n$ imply $y_n < H_n$, it is immediate that $\alpha_+(y', u) \subset \alpha_+(y, y', u)$. On the other hand, $u_n - u_{n+1} \leq 0$ and $y'_n = L_n$ imply $y_n \geq L_n$ and $(A(u - u'(y', u)))_n = 0$. Thus, $\alpha_-(y', u) \subset \alpha_-(y, y', u)$.

We can now prove (17). We rewrite (16) as

$$\begin{aligned} (A(u - u'(y', u)))_n &= (A(u - u^H(y', u)))_n \\ &\quad + (A(u^H(y', u) - u^L(y', u^H(y', u))))_n. \end{aligned} \quad (18)$$

From the definition of $u^H(y', u)$, we can easily conclude

- $(A(u - u^H(y', u)))_n = 0$ if $n \notin \alpha_+(y', u)$ and $u_n - u_{n+1} \geq 0$; and
- $(A(u - u^H(y', u)))_n \leq 0$ if $n \notin \alpha_+(y', u)$ and $u_n - u_{n+1} \leq 0$.

Therefore,

$$(A(u - u^H(y', u)))_n \leq 0, \quad \forall n \notin \alpha_+(y', u). \quad (19)$$

Moreover, by (16), (11), and Lemma 1, we have

$$\sum_{n=1}^N (A(u - u^H(y', u)))_n = u_1 - u_n^H(y', u) \geq 0. \quad (20)$$

Inequalities (19) and (20) together with the fact that

$\alpha_+(y', u) \subset \alpha_+(y, y', u)$ allow us to conclude

$$\begin{aligned} \sum_{n \in \alpha_-(y, y', u)} (A(u - u^H(y', u)))_n &= \sum_{n=1}^N (A(u - u^H(y', u)))_n \\ &\quad - \sum_{n \notin \alpha_-(y, y', u)} (A(u - u^H(y', u)))_n \\ &\geq 0. \end{aligned} \quad (21)$$

Similarly, the following is verified easily from the definition of $u^L(y', u^H(y', u))$:

— For $n \notin \alpha_-(y', u^H(y', u))$ and $u_{n+1}^H(y', u) - u_n^H(y', u) \geq 0$,

$$(A(u^H(y', u) - u^L(y', u^H(y', u))))_n = 0.$$

— For $n \notin \alpha_-(y', u^H(y', u))$ and $u_{n+1}^H(y', u) - u_n^H(y', u) \leq 0$,

$$(A(u^H(y', u) - u^L(y', u^H(y', u))))_n \geq 0.$$

These imply

$$(A(u^H(y', u) - u^L(y', u^H(y', u))))_n \geq 0, \quad \forall n \notin \alpha_-(y', u^H(y', u)). \quad (22)$$

Next, we show that

$$\alpha_-(y', u^H(y', u)) = \alpha_-(y', u). \quad (23)$$

Indeed, if $n \in \alpha_-(y', u)$, then from (10) and (11) we have $y'_n = L_n$, $u_n - u_{n+1} \leq 0$, and

$$u_n^H(y', u) = [u_n - u_{n+1} + u_{n+1}^H(y', u)]^+ \leq u_{n+1}^H(y', u).$$

From (10), we then conclude that $n \in \alpha_-(y', u^H(y', u))$, and thus $\alpha_-(y', u) \subset \alpha_-(y', u^H(y', u))$. Similarly, for any $n \in \alpha_-(y', u^H(y', u))$, we have $y'_n = L_n$ and $u_n^H(y', u) - u_{n+1}^H(y', u) \leq 0$. But from the definition of $u^H(y', u)$, we can easily see that the inequality is only possible when $u_n - u_{n+1} \leq 0$, and as a result, $n \in \alpha_-(y', u)$. Thus (23) is proved.

From (23) and the fact $\alpha_-(y', u) \subset \alpha_-(y, y', u)$ proved earlier, we have that if $n \in \alpha_-(y, y', u)$, then $n \notin \alpha_-(y', u^H(y', u))$. Thus, from (22) we have

$$\sum_{n \in \alpha_-(y, y', u)} (A(u^H(y', u) - u^L(y', u^H(y', u))))_n \geq 0. \quad (24)$$

Now (17) follows from (18), (21) and (24).

Similarly, we can show that $\sum_{i=1}^N [y_i(t) - y'_i(t)]^+$ is also nonincreasing.

For the general case, i.e., when $u(\cdot)$ is not necessarily piecewise-constant and right-continuous, the results can be proved through a limiting procedure. $\square$

So far we have obtained the transformation (13) that has allowed us to construct a modified deterministic control and the corresponding trajectory for any new initial condition, which does not stray too far away from the original trajectory. We shall now show that this is also the case for our stochastic problem when we use the construction procedure sequentially between the successive jumps of the Markov chain $\mathbf{k}(\cdot)$.

Proof of Main Lemma. For given $u(\cdot) \in \mathcal{A}(\mathbf{x}, \mathbf{k})$ and the new initial condition $\mathbf{x}' \in S$, we can construct the continuous process $\mathbf{x}'(t)$ satisfying

$$\frac{d\mathbf{x}'(t)}{dt} = A\mathbf{u}'(\mathbf{x}'(t), \mathbf{u}(t)), \quad \mathbf{x}'(0) = \mathbf{x}',$$

by applying the existence statement of Lemma 2 between successive jumps of the process $\mathbf{k}(\cdot)$. It follows from the above construction that the properties of controls and trajectories specified in Lemmas 1 and 2 hold also for the processes $\mathbf{x}(\cdot)$, $\mathbf{u}(\cdot)$, $\mathbf{x}'(\cdot)$ and $\mathbf{u}'(\cdot)$. This proves the statements (a) and (c) of Main Lemma.

In order to prove (b), we know from (1) that

$$\sum_{i=n}^N \frac{d}{dt}(x_i(t) - x'_i(t)) = u_n(t) - u'_n(t). \quad (25)$$

Integrating from 0 to t and using the results of Lemmas 1 and 2, we have

$$\begin{aligned}
 \int_0^t |u_n(s) - u'_n(s)| \, ds &= \int_0^t (u_n(s) - u'_n(s)) \, ds \\
 &= \sum_{i=n}^N (x_i(t) - x'_i(t)) - \sum_{i=n}^N (x_i - x'_i) \\
 &\leq \sum_{i=1}^N (x_i(t) - x'_i(t))^+ + \sum_{i=1}^N (x_i - x'_i)^- \\
 &\leq \sum_{i=1}^N (x_i - x'_i)^+ + \sum_{i=1}^N (x_i - x'_i)^- \\
 &= \sum_{i=1}^N |x_i - x'_i|.
 \end{aligned}$$

This proves the statement (b), and completes the proof of Main Lemma. $\square$

4. Concluding remarks

In this paper we have developed a theoretical framework for production planning in an N -machine flowshop with finite buffer capacities. We use the directional derivatives to describe the dynamics of the problem and the associated boundary condition. We prove the important property that the value function is locally Lipschitz continuous, and that it is a solution to a dynamic programming problem. The optimal feedback control is given in terms of the value function of the problem. The theoretical results characterizing the optimal control policies can be used in developing numerical approaches for solving production planning problems in dynamic stochastic flowshops. In view of verification part of Theorem 1, the problem is reduced to solving the DPEDD (6) to obtain the value function and find the optimal controls. In this connection, numerical schemes developed by Kushner and Dupuis (1992) can be used to compute the solution to (6).

Many important problems remain open. One of them is the problem in which machine breakdown rates depend on the

production rates as in Section 3.6 of Sethi and Zhang (1994). Another is the extension of our results to problems with the average cost criterion, since so far only singular or parallel machine problems have been dealt with in this context (see Sethi *et al.*, 1997).

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